

# Digital Material Passport

ID DMP-METAL-LINKS-001 - Version 1.0.0  
Issue Date 2025-05-14 - Certificate Type EN 10204 3.1

| Manufacturer                                                                                                                                                    |                                    | Customer                                                                                                                                                                               | Business Transaction |                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ACME Metal Works GmbH</b><br>Industrial Park 123<br>52066 Aachen<br>DE<br><a href="mailto:quality@acme-metal.example.com">quality@acme-metal.example.com</a> |                                    | <b>Engineering Solutions Ltd.</b><br>Tech Park Way 45<br>Cardiff<br>CF14 5DU<br>GB<br><a href="mailto:procurement@engisolutions.example.com">procurement@engisolutions.example.com</a> | <b>Order</b>         | PO-78901 · <b>5000 kg</b><br>Pos. 10 · 2025-04-20                                                                                                                   |
|                                                                                                                                                                 |                                    |                                                                                                                                                                                        | <b>Delivery</b>      | DN-56789 · <b>5000 kg</b><br>Pos. 1 · 2025-05-12<br>Internal Order ID: SO-12345                                                                                     |
| Product                                                                                                                                                         |                                    |                                                                                                                                                                                        |                      |                                                                                                                                                                     |
| Product Name                                                                                                                                                    | Structural Steel S355J2            |                                                                                                                                                                                        | Name (EN)            | 1.0577                                                                                                                                                              |
| Batch ID                                                                                                                                                        | H-10987-02                         |                                                                                                                                                                                        | Shape                | RoundBar - D 50 mm - Length 6000 mm                                                                                                                                 |
| Surface Condition                                                                                                                                               | Hot-rolled                         |                                                                                                                                                                                        | Packaging            | Hexagonal bundles of 5 pieces with wire binding in good condition.                                                                                                  |
| Weight                                                                                                                                                          | 5000 kg                            |                                                                                                                                                                                        |                      |                                                                                                                                                                     |
| Production Date                                                                                                                                                 | 2025-05-09                         |                                                                                                                                                                                        | Product Norm         | EN 10025-2 (2019)                                                                                                                                                   |
| Country of Origin                                                                                                                                               | DE                                 |                                                                                                                                                                                        | HS Code              | 720839 - Flat-rolled products of iron or non-alloy steel, of a width of 600 mm or more, hot-rolled, not clad, plated or coated                                      |
| Specification                                                                                                                                                   | EN 10025-2 - Rev 2019 - 2019-11-01 |                                                                                                                                                                                        | CN8 (EU)             | 72083900 - Flat-rolled products of iron or non-alloy steel, of a width of 600 mm or more, hot-rolled, not clad, plated or coated, of a thickness of 4.75 mm or more |

## Heat Treatment

| Process: Normalizing - Lot: HT-2024-11-15-B47 - Furnace: FURNACE-03 - Date: 2024-11-15 |             |          |         |            |
|----------------------------------------------------------------------------------------|-------------|----------|---------|------------|
| Stage                                                                                  | Temperature | Duration | Cooling | Atmosphere |
| Austenitizing                                                                          | 920 C       |          |         |            |

## Chemical Analysis

| Heat Number H-10987 - Melting Process EAF+LF - Casting Method ContinuousCasting - Casting Date 2025-05-08 - Sample Location Ladle |      |      |      |       |       |       |      |
|-----------------------------------------------------------------------------------------------------------------------------------|------|------|------|-------|-------|-------|------|
| Symbol                                                                                                                            | C    | Mn   | Si   | P     | S     | N     | CEV  |
| Unit                                                                                                                              | %    | %    | %    | %     | %     | %     | %    |
| Min                                                                                                                               |      |      |      |       |       |       |      |
| Max                                                                                                                               | 0.20 | 1.60 | 0.50 | 0.025 | 0.020 | 0.009 | 0.45 |
| Actual                                                                                                                            | 0.18 | 1.45 | 0.25 | 0.018 | 0.012 | 0.006 | 0.42 |
| CEV = C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15: 0.42%                                                                                        |      |      |      |       |       |       |      |

## Mechanical Properties

| Tensile Strength <sup>1</sup>          | Symbol | 1    | 2   | 3    | Average | Min  | Max  | Spec Min | Spec Max | Method        |
|----------------------------------------|--------|------|-----|------|---------|------|------|----------|----------|---------------|
| Value [MPa]                            | Rm     | 508  | 510 | 512  | 510     | 508  | 512  | 470      | 630      | EN ISO 6892-1 |
|                                        |        |      |     |      |         |      |      |          |          |               |
| Yield Strength <sup>1</sup>            | Symbol | 1    | 2   | 3    | Average | Min  | Max  | Spec Min | Spec Max | Method        |
| Value [MPa]                            | ReH    | 378  | 380 | 382  | 380     | 378  | 382  | 355      | 630      | EN ISO 6892-1 |
|                                        |        |      |     |      |         |      |      |          |          |               |
| Elongation after fracture <sup>1</sup> | Symbol | 1    | 2   | 3    | Average | Min  | Max  | Std Dev  | Spec Min | Method        |
| Value [%]                              | A      | 21.5 | 22  | 22.5 | 22      | 21.5 | 22.5 | 0.5      | 20       | EN ISO 6892-1 |

<sup>1</sup> 3 specimens tested

## Validation

We hereby certify that the material described above has been manufactured and tested in accordance with the requirements of EN 10204:2004 type 3.1 and the specified standards. The results comply with the requirements. The notarization can be verified at [app.s1seven.com](https://app.s1seven.com). The data schema is published at <https://schemas.materialidentity.org/metals-schemas/v0.1.1/schema.json>. Questions (e.g. on EN 10204 3.1) go to [quality@example.com](mailto:quality@example.com).

Validated by **Johann Weber**, Quality Inspector, Quality Assurance - 2025-05-14